

SEGMENT 02

The Sarvam Stack, after Epoch 2026

Five layers, twelve new products, and the gap between announced and callable

Epoch 2026 · Bengaluru · 30 July 2026 — ten days before this session

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One company, five layers

1

MODELS — what you call

Saaras (speech in) · Bulbul (speech out) · Sarvam-105B / 30B (reasoning) · Mayura & Sarvam-Translate (translation) · Sarvam Vision (documents)

2

APIs — the developer surface

REST · Batch · WebSocket streaming · OpenAI-compatible chat · Document AI jobs · Dubbing · Pronunciation dictionaries

3

PRODUCTS — what you compete with, partner with, or resell

Indus · Samvaad · Arya · Akshar · Studio · Kivi · Kaze · Sarvam Code · Edge

4

TOOLING — the productivity multiplier

Python & JS SDKs · MCP server · Agent Skills · llms.txt · Context7 · API Playground · Cookbook

5

DEPLOYMENT — where it runs

Sarvam Cloud · Private VPC · On-premise · Air-gapped · AWS Marketplace + SageMaker · Chanakya · Anvaya



Layer 1 — The models

Six model families, each doing one job well

The model line-up

Model	Job	Languages	Model ID	Status
Saaras v3	Speech to text, 5 modes	23 (22 Indic + En)	saaras:v3	Live
Bulbul v3	Text to speech, 30+ voices	11	bulbul:v3	Live
Sarvam-105B	Flagship reasoning LLM	11	sarvam-105b	Live
Sarvam-30B	Smaller reasoning LLM	11	sarvam-30b	Deprecated → use 105B
Mayura v1	Translation	11	mayura:v1	Live
Sarvam-Translate v1	Extended translation	23	sarvam-translate:v1	Live
Sarvam Vision 1.5	Document OCR + extraction	23	via doc_ai	Live
Saarika v2.5	Legacy ASR	11	saarika:v2.5	Legacy
Sarvam-M / 30B-16k	Superseded	—	—	Deprecated

★ Also on the chat endpoint, third-party: Gemma-4 31B and GLM 5.2. Note the pricing later — the Indic-native flagship is also the cheapest of the three.

Saaras — speech in



23 languages

All 22 constitutionally scheduled Indian languages plus English. The widest Indic ASR coverage available anywhere.



Five output modes

transcribe · translate · verbatim · translit · codemix. One audio file, five different jobs. This is the differentiator.



Three delivery paths

REST for clips under 30s · Batch for up to 20 files x 60 min with diarization · WebSocket for real-time with VAD.



Built for real audio

Code-mixed speech, heavy accent variance, and 8kHz telephony — the three things that break Western ASR models.



₹30/hour, or ₹45/hour with speaker diarization. Billed per second, rounded up per request.

Bulbul — speech out



30+ voices, 11 languages

Chosen by persona —
Conversational/Friendly,
News/Authoritative,
Entertainment/Dynamic — not by
personal preference.



Full control surface

Pitch, pace, loudness, tone, sample
rate. Plus pronunciation dictionaries for
brand names, SKUs and scheme names.



Streaming is mandatory

REST, HTTP stream, and WebSocket
with an end signal. For a voice agent,
first-audio latency is the entire
experience.



Telephony formats

mp3, linear16, mulaw, alaw, opus, flac,
aac, wav. mulaw and alaw are the ones
your phone system needs.



Bulbul v3 is ₹30/10K characters. Bulbul v2 is ₹15/10K. On a high-volume IVR, that single choice is your entire margin.

Sarvam-105B — the flagship

105B+

parameters, Mixture-of-Experts
with 128 sparse experts

12T

pre-training tokens across code,
math, multilingual and web

128K

context window,
Multi-head Latent Attention

Apache 2.0

open weights —
you are not locked in

Benchmark	Score	What it means
Math500	98.6	Multi-step mathematical reasoning
AIME 2025	88.3 (96.7 with tools)	Competition-level problem solving
BrowseComp	49.5	Agentic web navigation — matters for tool use
Tau2 (avg)	68.3	Tool use and environment interaction — highest among compared models
Indic pairwise	wins 90%	Against open and closed frontier models of its class

The language layer and Vision



Translation & script

- Mayura v1 — 11 languages, tuned quality, context preservation
- Sarvam-Translate v1 — 23 languages, open weights, wider coverage
- Both ₹20 per 10K characters — choose on coverage first, register second
- Registers: colloquial, modern, classical, formal
- Transliterate — script conversion preserving pronunciation, ₹20/10K
- Language ID — returns language AND script, ₹3.50/10K. Cheap enough to run on every inbound message as a router



Sarvam Vision 1.5 — documents

- 3B-parameter vision-language model, 23 languages
- Digitise — full OCR, layout and reading order preserved, tables to HTML/Markdown
- Extract — you define a JSON schema, it returns those fields
- Handles typed, scanned and handwritten Indic documents
- ₹0.50 per page — down from ₹1.50 in May 2026
- Hard limits: 10 pages per job, 200MB, 10 requests/minute on every plan



Epoch 2026

Twelve products in a single day — 30 July 2026

TEN DAYS AGO

What shipped at Epoch



Saaras V4

Speech recognition across all 22 scheduled languages. Plus V4 Multi-Speaker with diarisation for multi-party conversations.



Bulbul V4

More expressive, more human-sounding text to speech.



Vision 2.0

Digitising complex documents — an upgrade on the 1.5 generation.



Indus

One platform, six tools: work, voice, content, doc and coding agents, plus the inference stack.



Kivi

Desktop voice tool for speech-driven control and coding. Built on Saaras V4, natively multilingual.



Kaze

AI smart glasses that understand what you see and assist with everyday tasks.



Sarvam Code (Beta)

A coding assistant built on open models, to help developers write and understand code.



Sarvam Inference

India-hosted inference platform serving Sarvam-105B alongside frontier open models.



Chanakya & Anvaya

A complete AI system that runs entirely inside your own servers, and a custom 30B model — both for defence, intelligence and government.



Samvaad — now GA

Managed voice agents generally available at ₹3.50 per minute. Your build-versus-buy benchmark for the rest of today.



Instant phone numbers

Rent an Indian number in about 30 seconds with PAN and Aadhaar verification. Removes the biggest friction in shipping a voice product.



A 1 trillion+ model

Announced as in training from scratch, targeted live within six months. For coding, cybersecurity, scientific research and simulation.

Announced is not the same as callable

Announced at Epoch (30 July)	What the public API serves today	What you should do
Saaras V4 / V4 Multi-Speaker	saaras:v3	Build on v3. Watch the changelog.
Bulbul V4	bulbul:v3	Build on v3.
Vision 2.0	Sarvam Vision 1.5	Build on 1.5 via doc_ai.
Sarvam Inference	Not in public docs	Watch for access.
Kivi / Kaze	Product downloads, not APIs	Use as products.
Indus (6 tools)	Platform at indus.sarvam.ai	Use the platform directly.
Chanakya / Anvaya	Enterprise / government sales	Contact sales.
1T+ parameter model	Announced, ~6 months out	Nothing to do yet.



Say this out loud when you teach it: everything we run today is on what is actually callable. Being the person who separates announced from shipped buys more trust than any demo.



Layer 3 — The products

What you compete with, partner with, or resell

The product portfolio

Product	What it is	Why a developer cares
Indus	Agentic platform — six tools in one	The reference architecture for what you'd build
Samvaad	Conversational AI — voice, WhatsApp, web	GA at ₹3.5/min. Your build-vs-buy benchmark
Arya	Enterprise agent platform	Teaches the patterns your own agents need
Akshar	Document digitisation	Vision, productised
Studio	Content transformation and dubbing	Media and edtech vertical
Kivi	Desktop voice tool	A product category wide open in your vertical
Kaze	AI smart glasses	Ambient computing, early
Sarvam Code	Coding agents (Beta)	Dev tooling surface
Sarvam Edge	On-device inference	Offline and rural, low-connectivity
Chanakya / Anvaya	On-prem and defence stack	How regulated buyers actually deploy
Model Training	Custom model training service	The enterprise upsell path

Arya — the patterns worth stealing



Checkpointed steps

A failure at step 40 does not lose the first 39. Recovery is instant. 50-step workflows just work.



Total observability

Every reasoning step, tool call and decision is logged and inspectable. Replay any step with different inputs.



Scoped permissions

Connect private tools, databases and APIs without exposing them. Each agent accesses only what it needs.



Model portability

Bring your own model or use theirs. Swap vendors without rewriting workflows. A/B test across providers.



Self-improving

Structured feedback loops and persistent memory — agents learn from corrections and carry it forward.



Long-run reliable

Workflows spanning hours or days. State managed across long-horizon processes without silent failure.



You are not being sold Arya. You are learning the checklist an enterprise buyer will hold you to — and 'LLMs reason, code executes' is the design law underneath all of it.



Layer 4 — Developer tooling

The multiplier most people skip

Four ways to make your AI assistant good at this stack

1

Official SDKs — `pip install sarvamai / npm install sarvamai`

Typed clients with async, retries and streaming built in. Reduces integration from hours to minutes.

2

MCP server — `uvx sarvam-mcp`

Exposes every public Sarvam API as a tool to Claude Desktop, Claude Code, Cursor, Windsurf, Zed. Two namespaces: `sarvam_tools_*` to call APIs, `sarvam_code_*` for docs and snippets.

3

Agent Skills — `npx skills add sarvamai/skills`

Six skills (chat, speech-to-text, text-to-speech, translate, voice-agents, vibe-coding) that teach your assistant the SDK quirks it otherwise gets wrong.

4

llms.txt and Markdown docs

Append `.md` to any docs URL. `/llms.txt` for the index, `/llms-full.txt` for the whole corpus — built for RAG ingestion and one-shot context.



Without the skills installed, your assistant writes `client.chat.completions.create(...)`. The real method is `client.chat.completions(...)`. We will prove this live in three minutes.

Which tool, when

	Agent Skills	MCP server	llms.txt
When it loads	Once, installed into your project or editor	Live, queried on demand while coding	Fetches manually when you need it
Best for	Baking in SDK corrections an assistant keeps getting wrong	Interactive sessions needing current API details, or to call Sarvam directly	Bulk ingestion, RAG indexes, one-shot context
Setup	npx skills add sarvamai/skills	One-time MCP client config	None — just a URL

★ They are complementary. A skill stops the wrong method name; the MCP server and llms.txt supply the parameter tables a skill deliberately leaves out.

E

Layer 5 — Where it runs

Managed, private, on-premise, air-gapped

Four deployment postures

1

Sarvam Cloud

Fully managed, automatic scaling, fastest time to value. Pay per call. Data transits Sarvam. Where you start.

2

Private Cloud (VPC)

Your security perimeter, their management. The middle option most mid-size enterprises land on.

3

On-premise / SageMaker

Saaras v3, Bulbul v3 and Vision self-hosted via AWS Marketplace. Audio and documents never leave your VPC.

4

Air-gapped

Network isolation enabled, no outbound internet from the model container. Chanakya for defence and national security.

₹

The crossover calculation

Per-call pricing wins until volume justifies a permanently-running GPU. At ₹30/hour for STT, work out how many audio-hours a month justify a dedicated endpoint. That number is the most useful thing you can take to a bank.

★

No licence key, no proxy

AWS Marketplace handles entitlement. Subscribe, deploy via console, boto3 or Terraform, and the endpoint is licensed for the instances you run.

The integrations you will actually reach for



LiveKit

WebRTC-native voice agents. Production-hardened. Best for web and app. First-party Sarvam plugin plus a production best-practices guide.



Pipecat

Python-native pipeline. Easier to reason about. Best for learning and custom flows. Also has a production guide.



Twilio / Exotel / Vapi

Telephony. Exotel is usually right for India — cheaper, better DLT/TRAI compliance. Or rent a number from Sarvam in 30 seconds.



n8n

Workflow automation. Call Sarvam APIs as drop-in nodes, no code required.



LangChain

Agent scaffolding. Works via the OpenAI-compatible endpoint with a base-URL swap.



Vercel AI SDK

Streaming UI for JavaScript and TypeScript apps. Same base-URL swap story.



Because chat completions are OpenAI-compatible, most existing code works with a base-URL change. We will demo that swap in the next segment — it takes ninety seconds.

Language coverage — check before you pick a model

23 23 languages — Saaras v3, Sarvam-Translate, Vision

- Hindi hi-IN · Bengali bn-IN · Tamil ta-IN · Telugu te-IN
- Marathi mr-IN · Gujarati gu-IN · Kannada kn-IN · Malayalam ml-IN
- Odia od-IN · Punjabi pa-IN · Assamese as-IN · Urdu ur-IN
- Nepali ne-IN · Konkani kok-IN · Kashmiri ks-IN · Sindhi sd-IN
- Sanskrit sa-IN · Santali sat-IN · Manipuri mni-IN · Bodo brx-IN
- Maithili mai-IN · Dogri doi-IN · English en-IN

11 11 languages — Bulbul v3, Mayura, Sarvam-105B

- Hindi hi-IN · Bengali bn-IN · Tamil ta-IN · Telugu te-IN
- Gujarati gu-IN · Kannada kn-IN · Malayalam ml-IN
- Marathi mr-IN · Punjabi pa-IN · Odia od-IN · English en-IN
- This asymmetry matters: you can TRANSCRIBE Santali but you cannot SPEAK it back yet.
- Design your product around the narrowest model in your chain.

★ The narrowest model in your pipeline defines your product's language coverage. Check it before you promise a customer 22 languages.

The restaurant with a very long menu

A new restaurant opens with a menu of forty dishes. You order the biryani. The waiter says: "That one is coming soon."

You order the dosa. "Also coming soon."

Eventually you learn that eight dishes are actually available today, and the rest are on the menu because the chef intends to make them.

The menu is not lying. It is a roadmap printed as a menu.



Epoch announced twelve products. The API serves a subset today. Both facts are true, and the developer who knows which is which does not waste a weekend building against a model that is not there yet.

Where this is going

1T+

parameter model announced,
live within ~6 months

V4 / 2.0

speech and vision generations
rolling into the API

6

tools converging into
one Indus platform

₹3.5

per minute — managed voice
agents, generally available

- The direction is unmistakable: from APIs you assemble, toward platforms you deploy. Samvaad and Indus are that shift made concrete.
- For a developer this is not a threat — it is a signal. Platform companies move up-market. The vertical, the SMB and the workflow-adjacent stay open.
- The open-weights release of 30B and 105B under Apache 2.0 is the safety valve. If pricing or terms ever change, you can self-host. Architect for that from day one.

“

Anyone can read a launch blog.
Knowing what actually works today is the job.

— End of Segment 02. Next: we run code.